

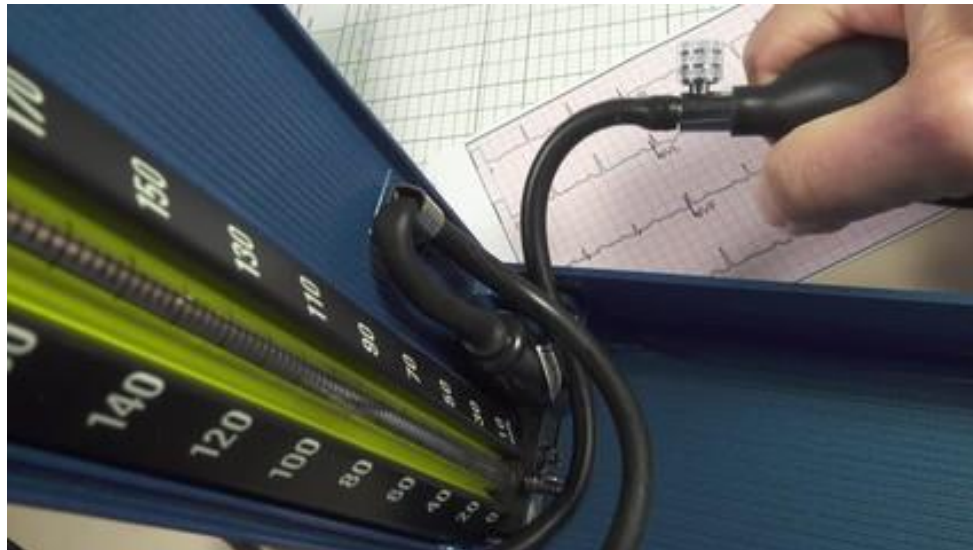
ADVANCED BIOSTATISTICS

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Outline

1. Concepts
2. Statistical tests for advanced biostatistics
3. Hands on
4. Reporting



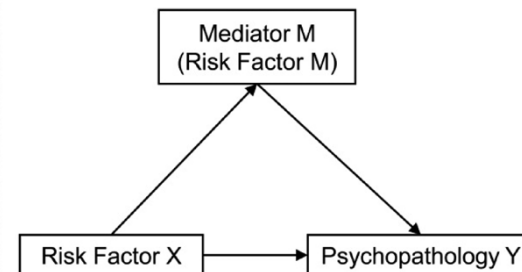
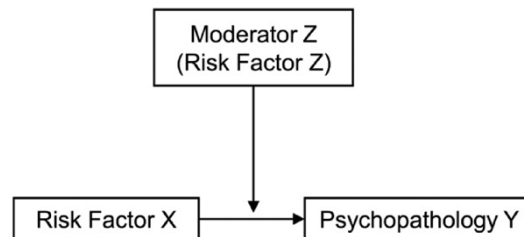
Multivariate concepts

(a) Direct Predictor

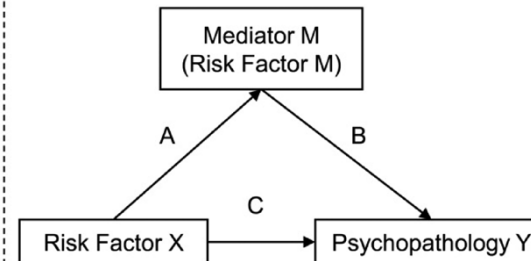
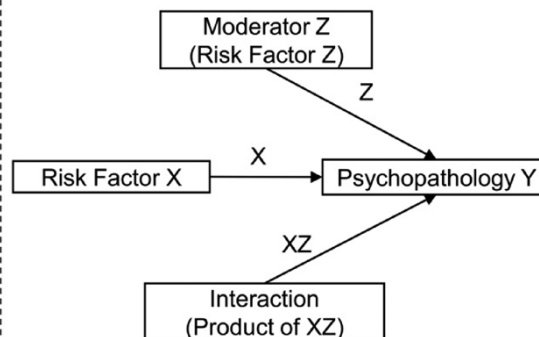
(b) Moderation

(c) Mediation

Conceptual models



Statistical models

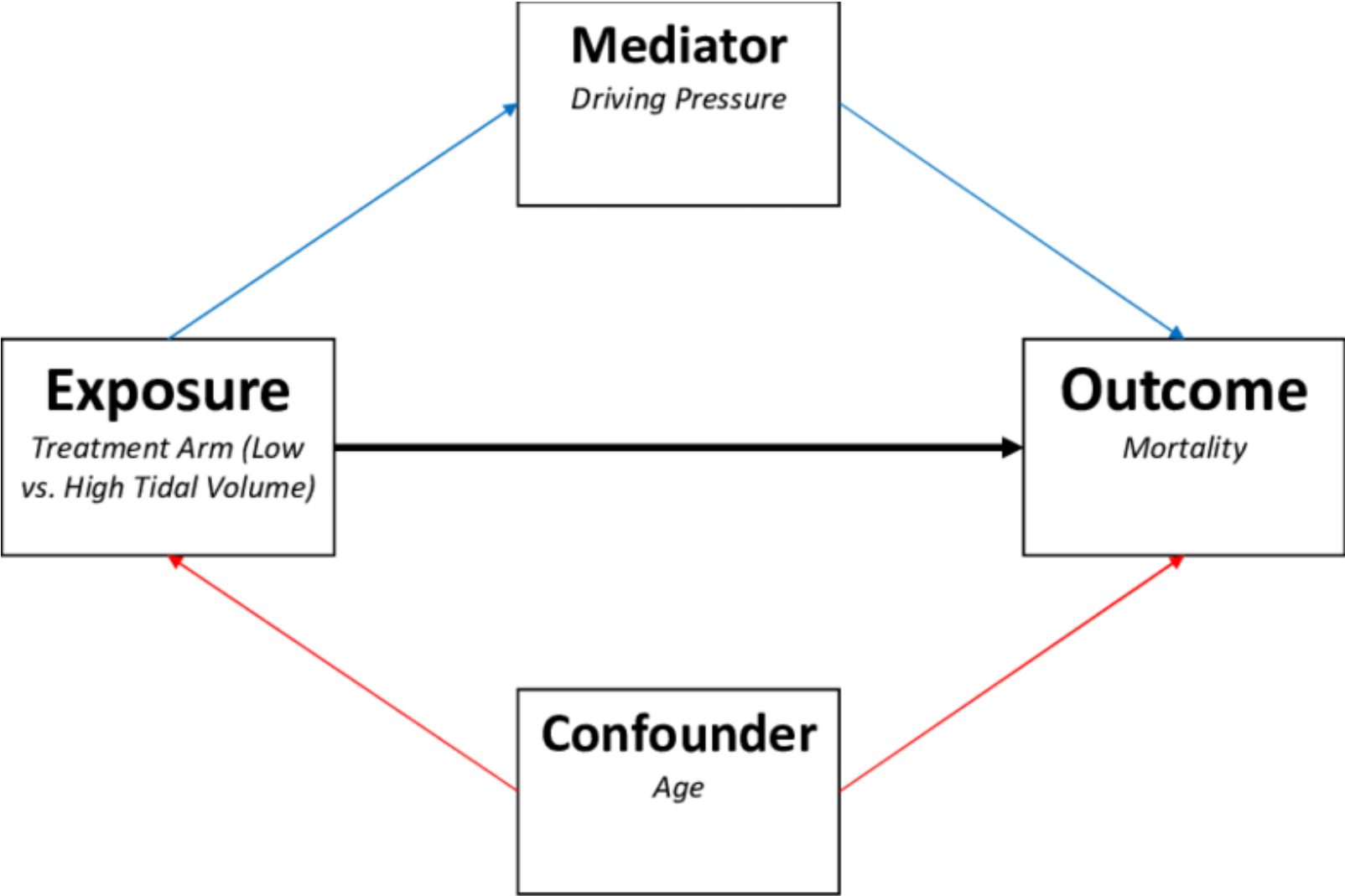


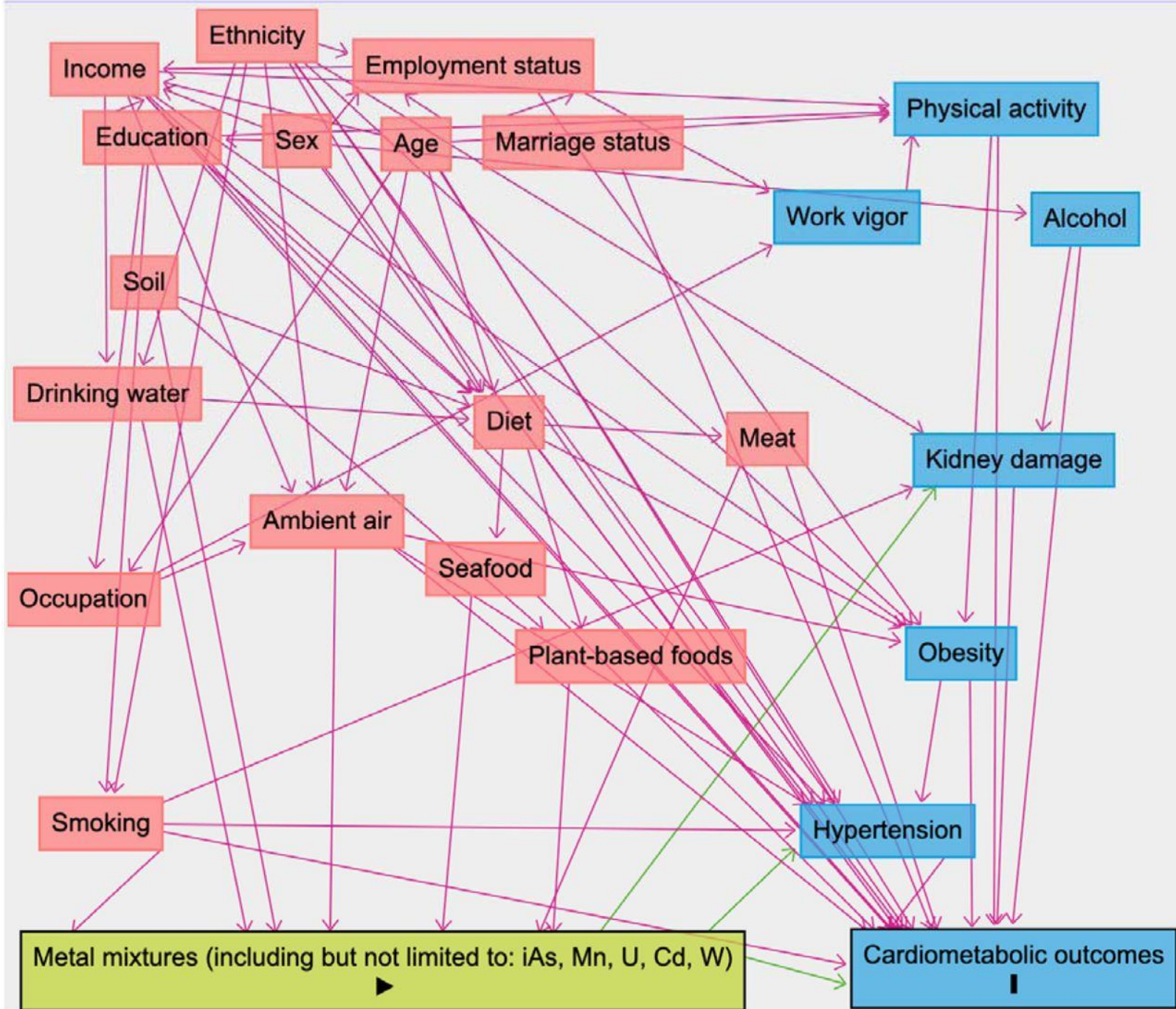
INDEPENDENT

- Factor/Exposure/Predictor
- Confounder
- Moderator
- Mediator
- Level of measurement
- Number of variables

DEPENDENT

- Outcome
- Level of measurement
 - Continuous or categorical
- Number of variables





Lifestyle factors

Smoking; Alcohol drinking;
Low physical activity; Low
fruits and vegetables intake,
etc.

Nutritional deficiency

Low folate;
Low Vitamin B12 and
B6

Genetic factors

CBS deficiency;
MTHRF deficiency

Disease

Kidney failure, etc.

Elevated homocysteine

Hcy $\geq$ 15 μ mol/L

Hyperhomocysteinemia

Hypertension

Synergistic effect

H-type hypertension

Oxidative stress status

Endothelial dysfunction

Atherosclerosis

**Increased risk of cardio/
cerebrovascular diseases**

Note:

CBS: Cystathionine beta-synthase

MTHRF: Methyltetrahydrofolate reductase

Hcy: Homocysteine

Why we do advanced analysis?

1. In real life, risks are multifactorial towards a particular disease
2. To get better and more precise estimate/magnitude of association
3. To control for confounder (nuisance)
4. To know whether the effect of independent variable on the outcome is same across different level of another variable (moderator)
5. To clarify a known relationship by exploring the mechanism an independent variable to the outcome through another variable (mediator)

Multivariate test

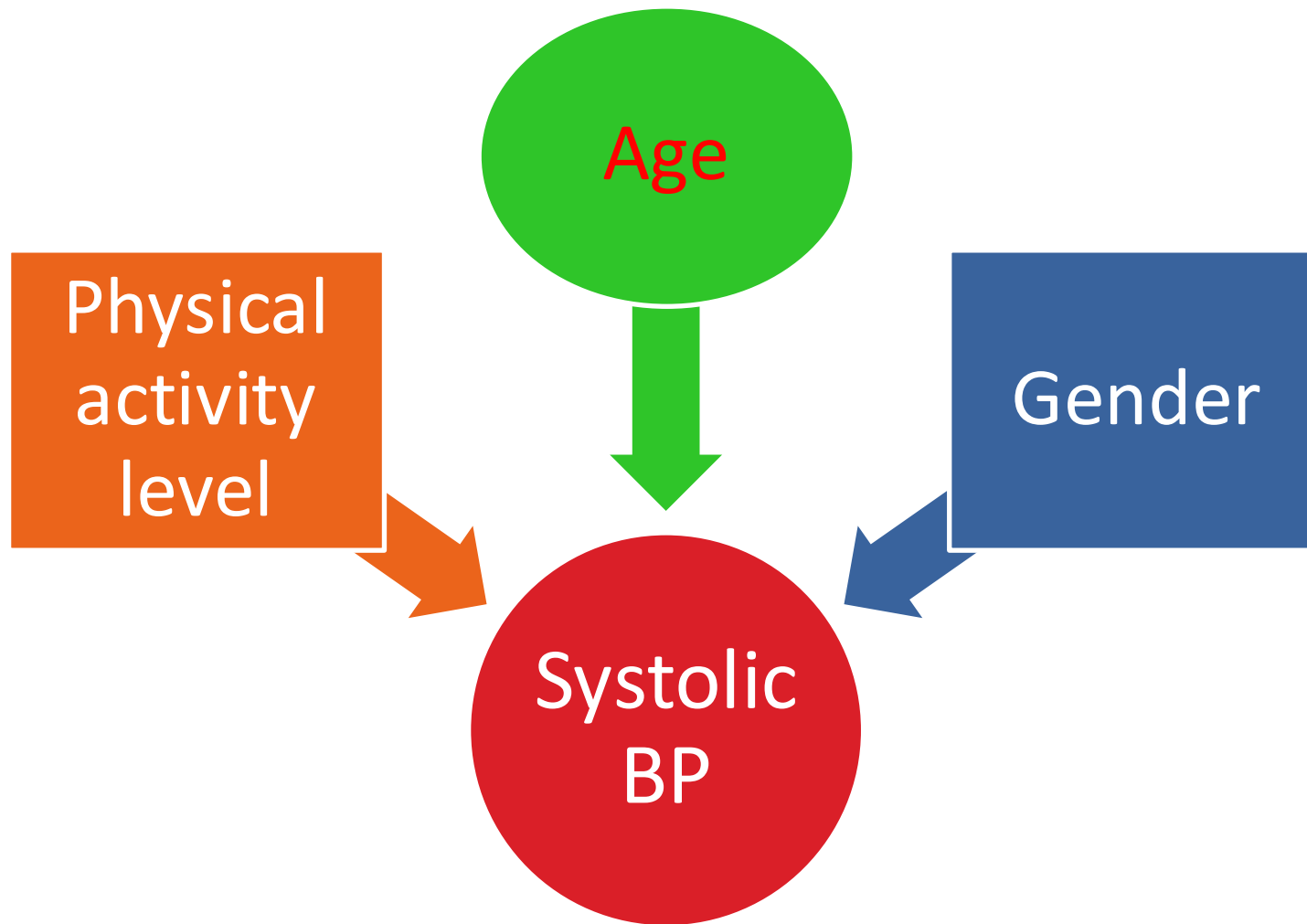
Dependent variable	Independent variable	Test
Numerical (normal)	Numerical (main) + categorical/numerical	Linear regression
Categorical (binary)	categorical (main) + Numerical/Categorical	Logistic regression Binary/Multinomial/Ordinal
Numerical (normal)	2 Categorical (≥ 2 categories)	Two-way ANOVA
Numerical (normal)	Covariate: Numerical IV: (categorical ≥ 2 cat)	ANCOVA
2 Numerical (normal)	Categorical (≥ 2 categories)	MANOVA
2 Numerical (normal)	Covariate: Numerical IV: (categorical ≥ 2 cat)	MANCOVA
Numerical (multiple times measured, ≥ 2 times)	Categorical (≥ 2 categories)	Repeated measure ANOVA
Numerical (multiple times ≥ 2 times) (Within)	Categorical (≥ 2 categories) + Categorical/Continuous (Between)	Mixed ANOVA

*Multivariate = >1 DV

*Multivariable = >1 IV

*Co variate = Continuous variable

Multiple linear regression



Multiple linear regression equation

$$Y = B_0 + B_1x_1 + B_2x_2$$

$$SBP = \text{constant} + PA + \text{Age} + \text{Gender}$$

Multiple linear regression

Assumptions

- i. Random sampling
 - Study design
- ii. Linearity
 - It is assumed that the relationship we are modelling is a linear one between the outcome and the main predictor.
- iii. Independence
 - It is assumed that all of the values of the dependent variable are independent (each value of the dependent variable comes from a separate entity)
- iv. Normality distributed error
 - It is assumed that the residuals in the model are random, normally distributed variables with a mean of 0.
- v. Equal variance (homoscedasticity)
 - At each level of the independent variable(s), the variance of residual terms should be constant.

Steps

- Scatter plot
- Correlation between main iv and dv (linearity)
- Simple linear regression (crude coefficient)
- Multiple linear regression (adjusted coefficient)
- Analyze > Regression > Linear
- Statistics
 - estimates, confidence intervals, model fit, R square, Descriptive

Reporting

Table 7.16 Sample Regression Table, With Confidence Intervals in Brackets

Table 4

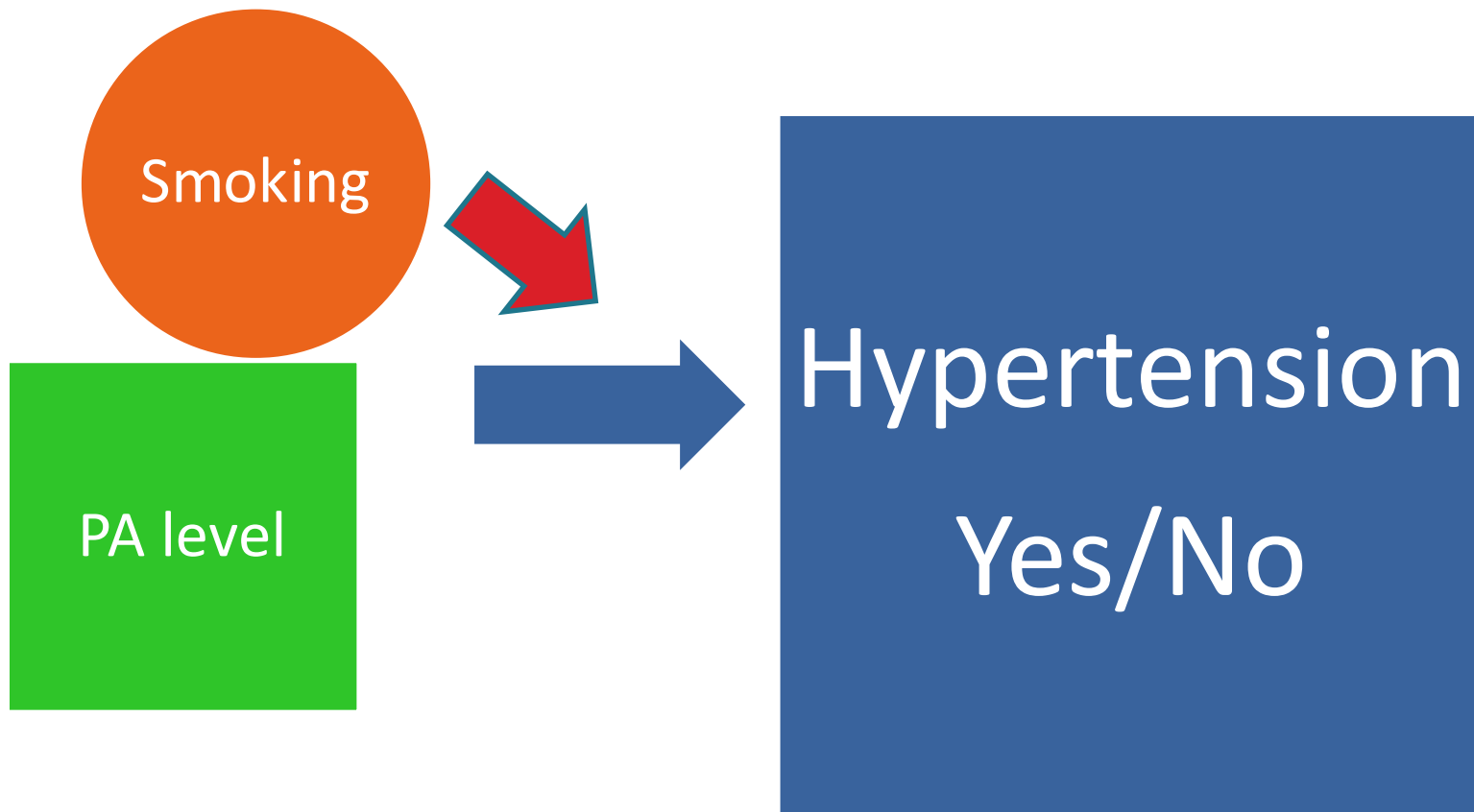
Regressions of Associations Between Marital Satisfaction and Average Levels of Marital Behavior

Variable	B	SE	t	p	95% CI
Angry behavior					
Actor					
H → H	−98.90	40.20	−2.46	.016	[−179.1, −18.7]
W → W	−87.11	30.87	−2.82	.006	[−148.7, −25.6]
Partner					
W → H	−76.18	39.43	−1.93	.057	[−154.8, 2.4]
H → W	−91.80	38.16	−2.41	.019	[−167.9, −15.7]
Disregard					
Actor					
H → H	−38.62	27.86	−1.39	.170	[−94.2, 16.9]
W → W	−47.54	26.99	−1.76	.082	[−101.4, 6.3]
Partner					
W → H	−82.81	32.01	−2.59	.012	[−146.6, −19.0]
H → W	−79.36	27.16	−2.92	.005	[−133.5, −25.2]
Distancing					
Actor					
H → H	−47.42	24.72	−1.92	.059	[−96.7, 1.9]
W → W	3.04	23.48	0.13	.897	[−43.8, 49.8]
Partner					
W → H	−0.05	23.91	0.00	.998	[−47.7, 47.6]
H → W	−53.50	24.47	−2.19	.032	[−102.3, −4.7]

square brackets
around confidence
intervals

Note. CI = confidence interval; H → H = husband-as-actor effect on the husband's own marital satisfaction; W → W = wife-as-actor effect on the wife's own marital satisfaction; W → H = wife-as-partner effect on the husband's satisfaction; H → W = husband-as-partner effect on the wife's satisfaction.

Multiple Logistic Regression



Multiple Logistic Regression Assumptions

1. DV dichotomous
2. > 1 IV (mixed numerical+dichotomous)
3. Independence of observation
4. Linear relationship between any continuous independent variables and the logit transformation of the dependent variable

Multiple Logistic Regression Equation

$$\text{Log (likelihood)} = B_0 + Bx_1 + Bx_2$$

$$\text{Log (likelihood of being hypertensive)} = \text{constant} + \text{OR(smoking)} + \text{OR(PA level)}$$

Steps

- Simple logistic regression (crude OR)
- Multiple logistic regression (adjusted OR)
- Analyze > Regression > Binary logistic
- Options-confidence intervals
- Categorical – reference category
- **Dummy variables**

Reporting

Table 7.16 Sample Regression Table, With Confidence Intervals in Brackets

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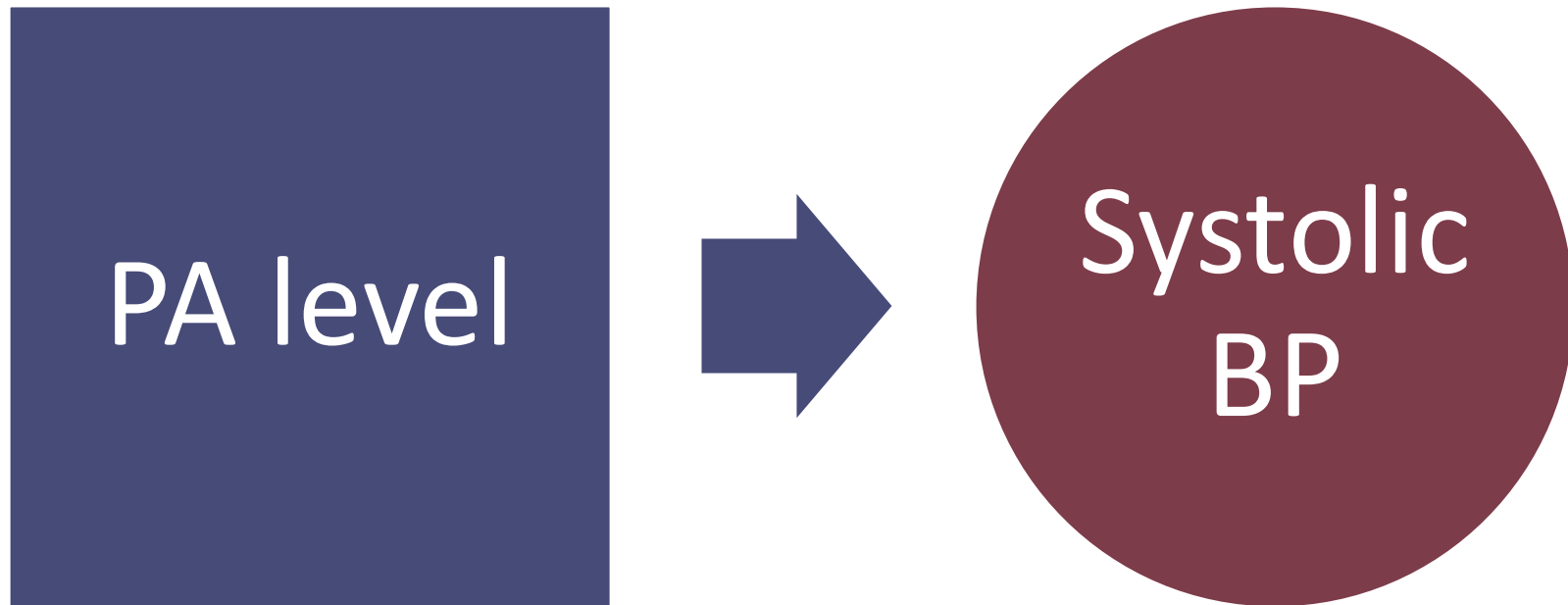
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square brackets
around confidence
intervals

Note. CI = confidence interval; H → H = husband-as-actor effect on the husband's own marital satisfaction; W → W = wife-as-actor effect on the wife's own marital satisfaction; W → H = wife-as-partner effect on the husband's satisfaction; H → W = husband-as-partner effect on the wife's satisfaction.

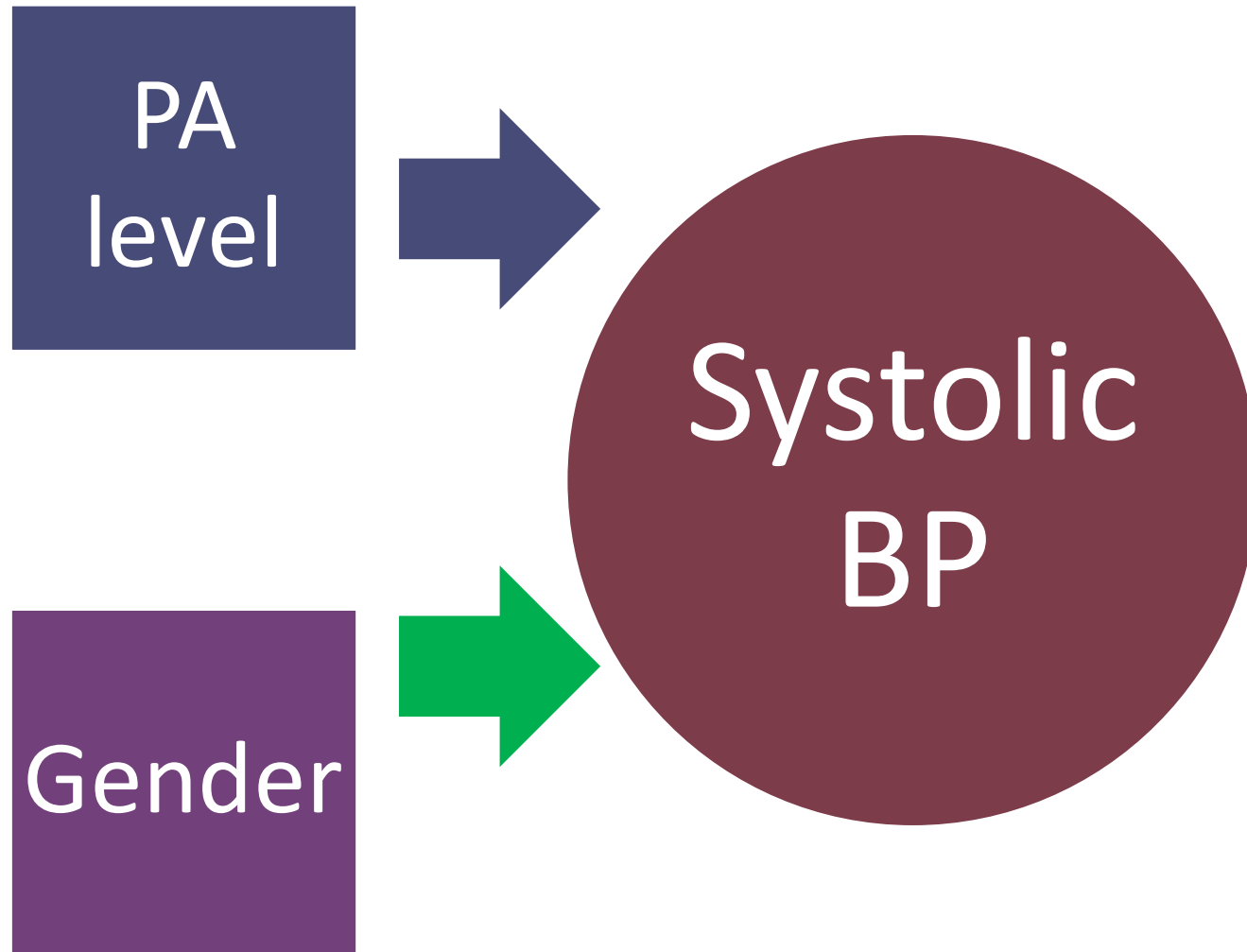
ANOVA



ANOVA Assumptions

- i. DV continuous
 - Normally distributed
- ii. One IV should have more than 2 categories
- iii. Independence
 - It is assumed that all of the values of the dependent variable are independent (each value of the dependent variable comes from a separate entity)
- iv. No significant outlier
 - Checked by sorting the data to look for illogical extreme values
- v. Equal variance (homoscedasticity)
 - Checked by levene's test

Two way ANOVA



Two way ANOVA

Assumptions

- i. DV continuous
 - Normally distributed
- ii. Two IV should have more than 2 categories
 - Independent of each other
- iii. Independence
 - It is assumed that all of the values of the dependent variable are independent (each value of the dependent variable comes from a separate entity)
- iv. No significant outlier
 - Checked by sorting the data to look for illogical extreme values
- v. Equal variance (homoscedasticity)
 - Checked by levene's test

Steps

- Analyze > General Linear Model > Univariate
- Enter Dependent Factor and Fixed factor
- Post hoc – Bonferroni or Tamhane's T2
- Option – Descriptives and homogeneity test

Reporting

Table 7.13 Sample Analysis of Variance Table (Option 2)

Table 2

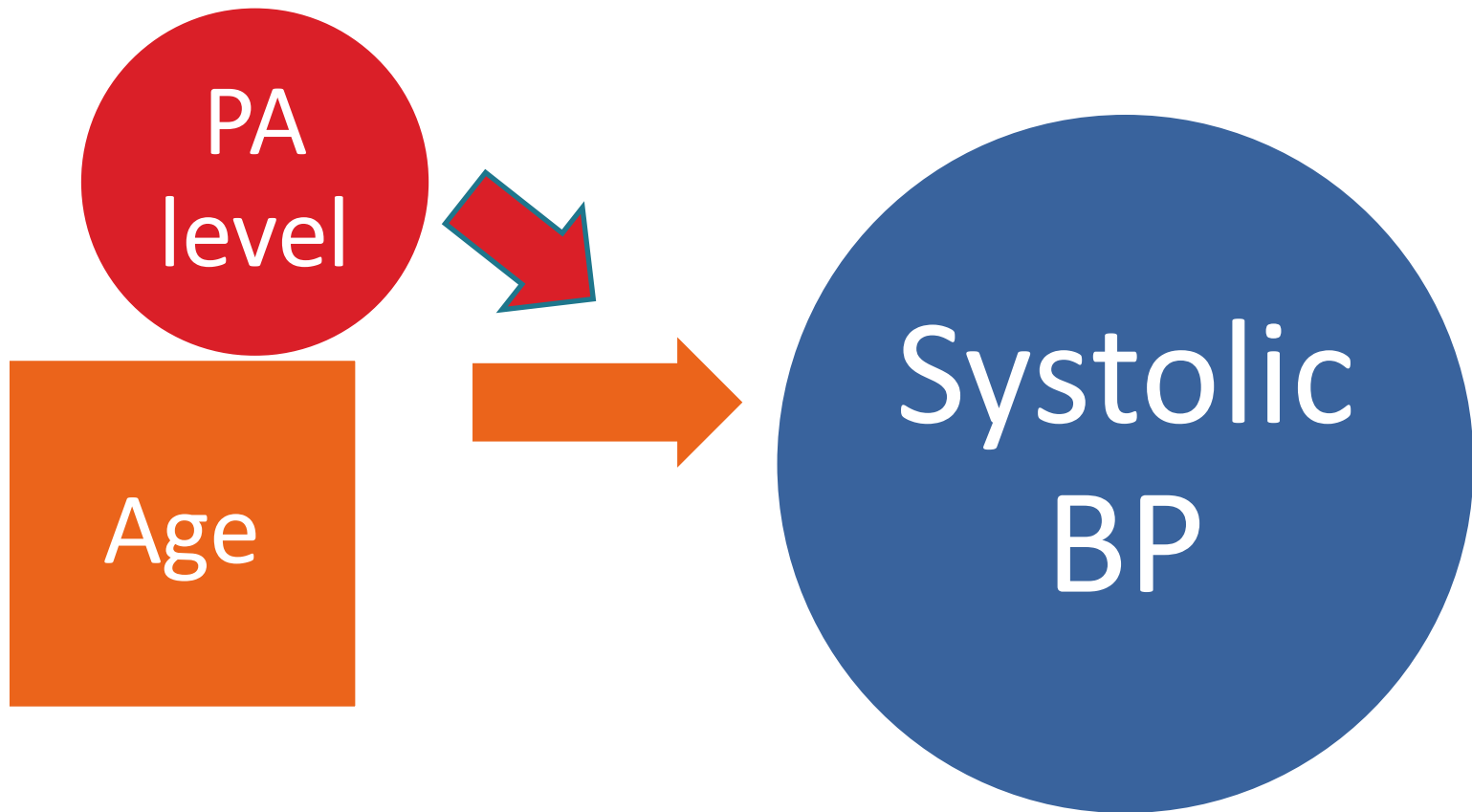
Means, Standard Deviations, and Two-Way ANOVA Statistics for Study Variables

Variable	SMT		Control		ANOVA			
	M	SD	M	SD	Effect	F ratio	df	η^2
Psychological strain								
Time 1	0.24	0.30	0.22	0.29	G	2.82	1,151	.02
Time 2	0.16	0.27	0.27	0.32	T	0.38	2,302	.00
Time 3	0.16	0.26	0.26	0.31	G $\times$ T	4.64**	2,302	.03
Emotional exhaustion								
Time 1	2.82	1.47	2.50	1.25	G	0.32	1,151	.00
Time 2	2.55	1.31	2.47	1.28	T	6.59**	2,302	.04
Time 3	2.36	1.39	2.43	1.16	G $\times$ T	3.89*	2,302	.03
Depersonalization								
Time 1	1.20	1.09	1.12	1.05	G	0.07	1,149	.00
Time 2	1.13	1.07	1.25	1.16	T	0.67	2,302	.00
Time 3	1.00	0.93	1.24	0.93	G $\times$ T	3.04*	2,302	.02

Note. N = 153. ANOVA = analysis of variance; SMT = stress management training group; Control = wait-list control group; G = group; T = time.

*p < .05. **p < .01.

ANCOVA



ANCOVA

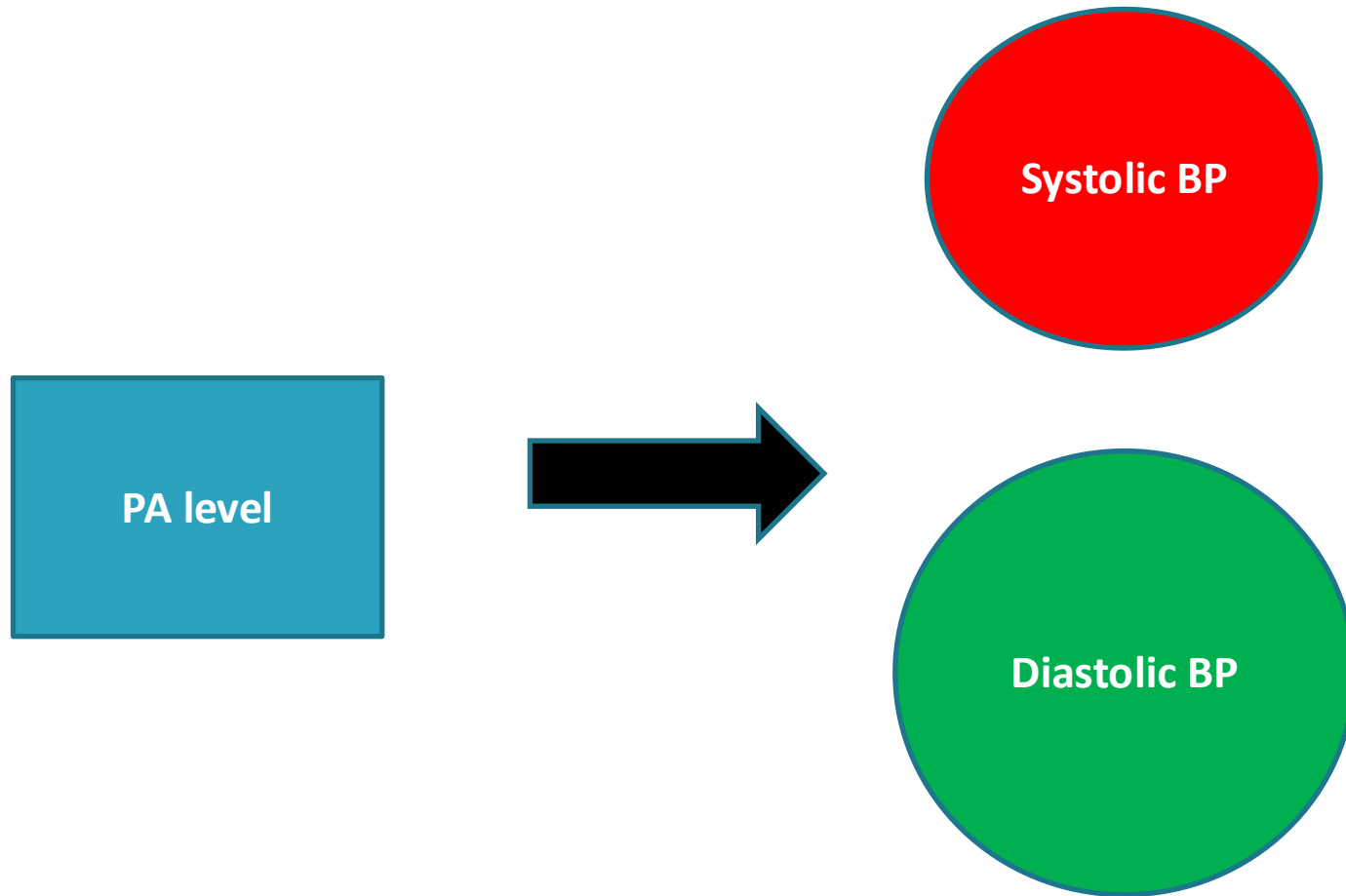
Assumptions

- i. DV and covariate continuous
- ii. IV should have more than 2 categories
 - Independent of each other
- iii. Independence
 - It is assumed that all of the values of the dependent variable are independent (each value of the dependent variable comes from a separate entity)
- iv. No significant outlier
 - Checked by sorting the data to look for illlogical extreme values
- v. Equal variance (homoscedasticity)
 - Checked by levene's test
- vi. Covariate should be linearly related to the dependent variable at each level of the independent variable
- vii. Homogeneity of regression slopes
- viii. Residuals should be approximately normally distributed for each category of the independent variable.

Steps

- Analyze > General Linear Model > Univariate
- Enter Dependant Factor , Fixed factor and covariate
- Post hoc – Bonferroni or Tamhane's T2
- Option – Descriptives and homogeneity test

MANOVA



MANOVA

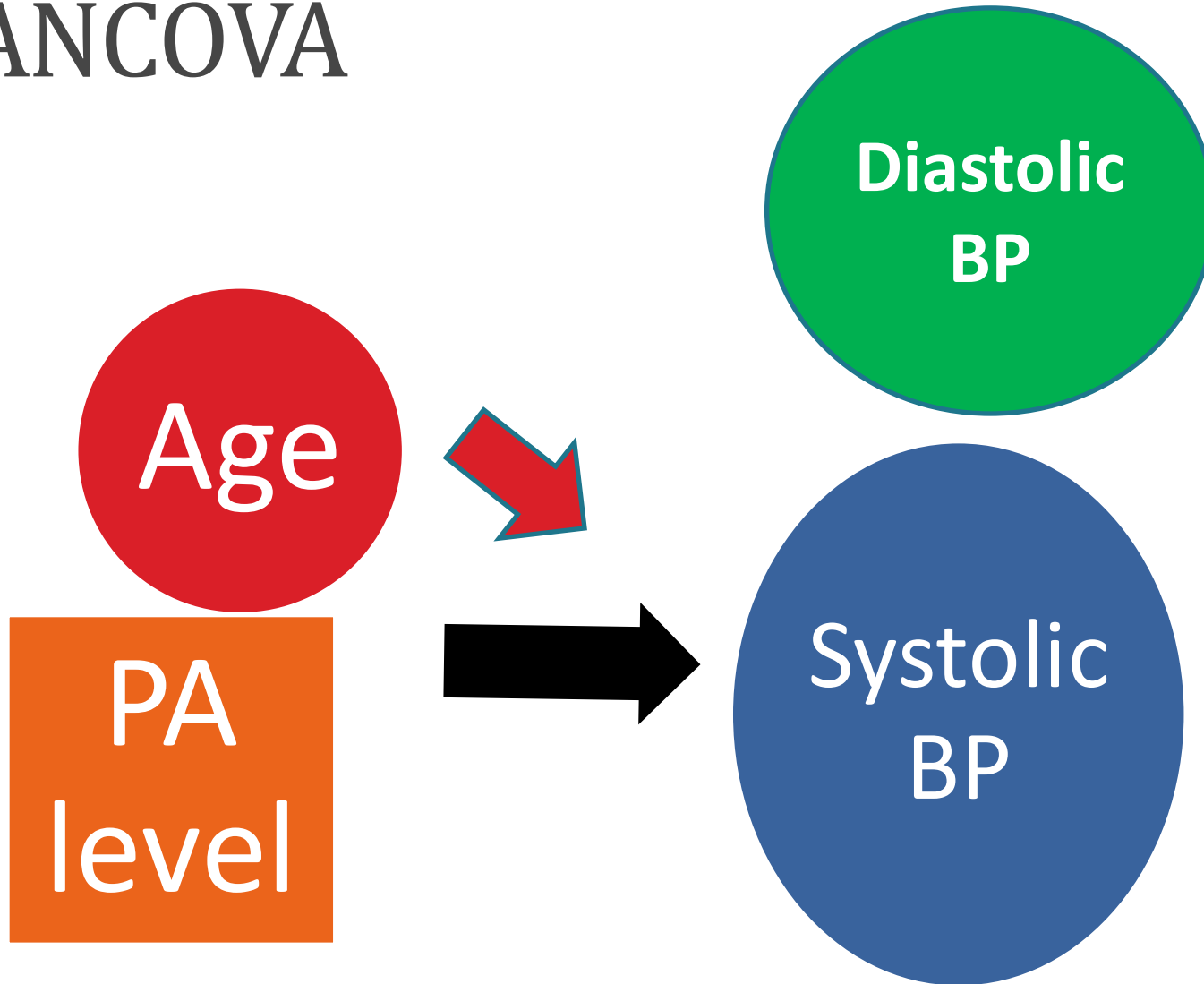
Assumptions

- i. Two DV numerical
- ii. IV should have more than 2 categories
- iii. Independence of observation
- iv. No significant outlier
- v. Equal variance (homoscedasticity)
- vi. Adequate sample size
- vii. Multivariate normality
- viii. No multicollinearity
- ix. Linear relationship between each pair of dependent variables for each group of the independent variable

Steps

- Check correlation between dependant variable
- Analyze > General Linear Model > Multivariate
- Enter dependant and fixed factor
- EM Means and post hoc - enter the main factor variable

MANCOVA



MANCOVA

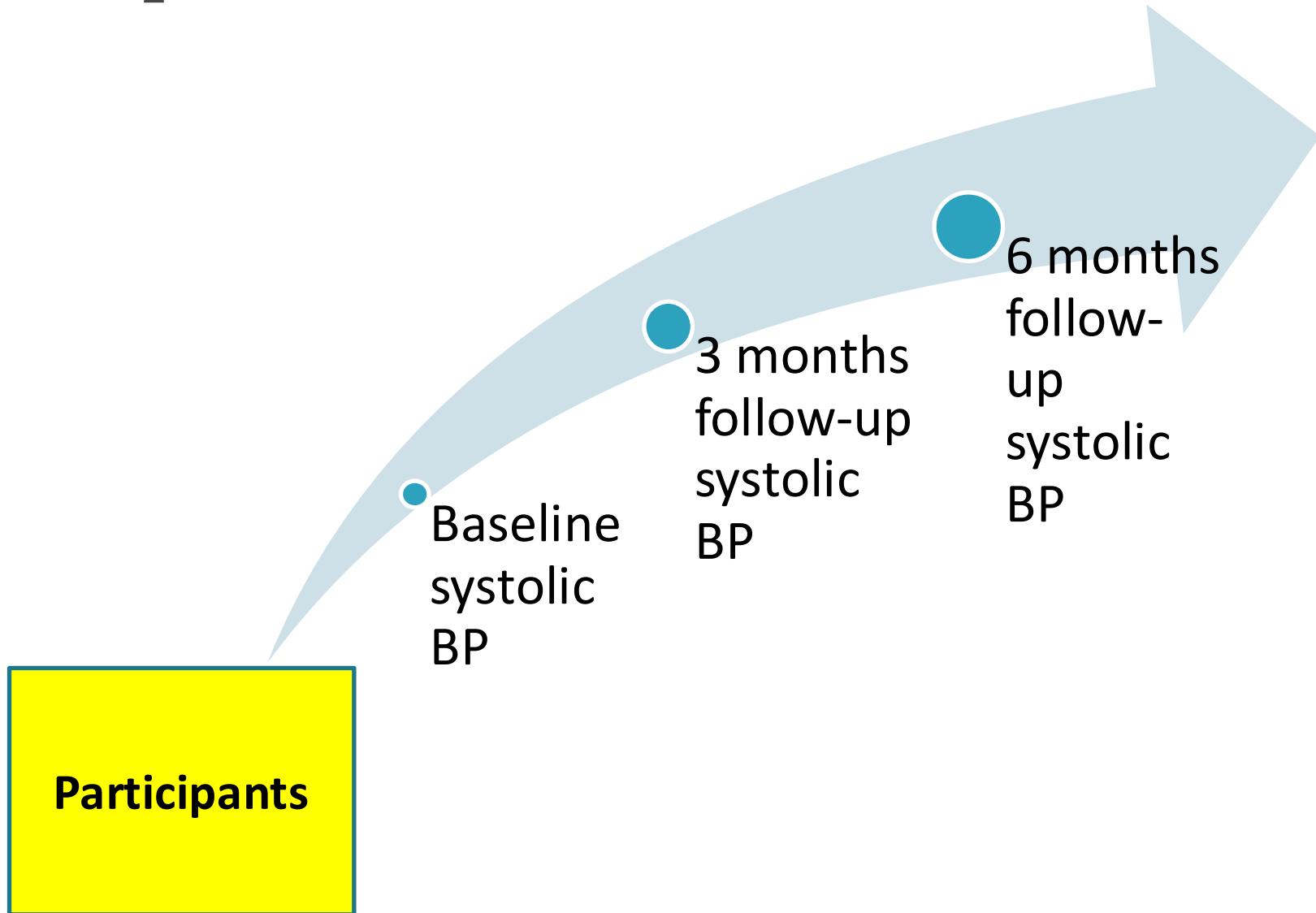
Assumptions

- i. Two DV numerical
- ii. IV should have more than 2 categories
- iii. Covariate numerical
- iv. Independence
- v. Linear relationship between each pair of dependent variables for each group of the independent variable
- vi. Linear relationship between the covariate and each dependent variable within each group of the independent variable
- vii. Homogeneity of regression slopes
- viii. Homogeneity of variances and covariances (Box M)
- ix. No significant outliers
- x. Multivariate normality

Steps

- Check correlation between dependant variable
- Analyze > General Linear Model > Multivariate
- Enter dependant , fixed factor and covariate
- EM Means and post hoc - enter the main factor variable

Repeated Measure ANOVA

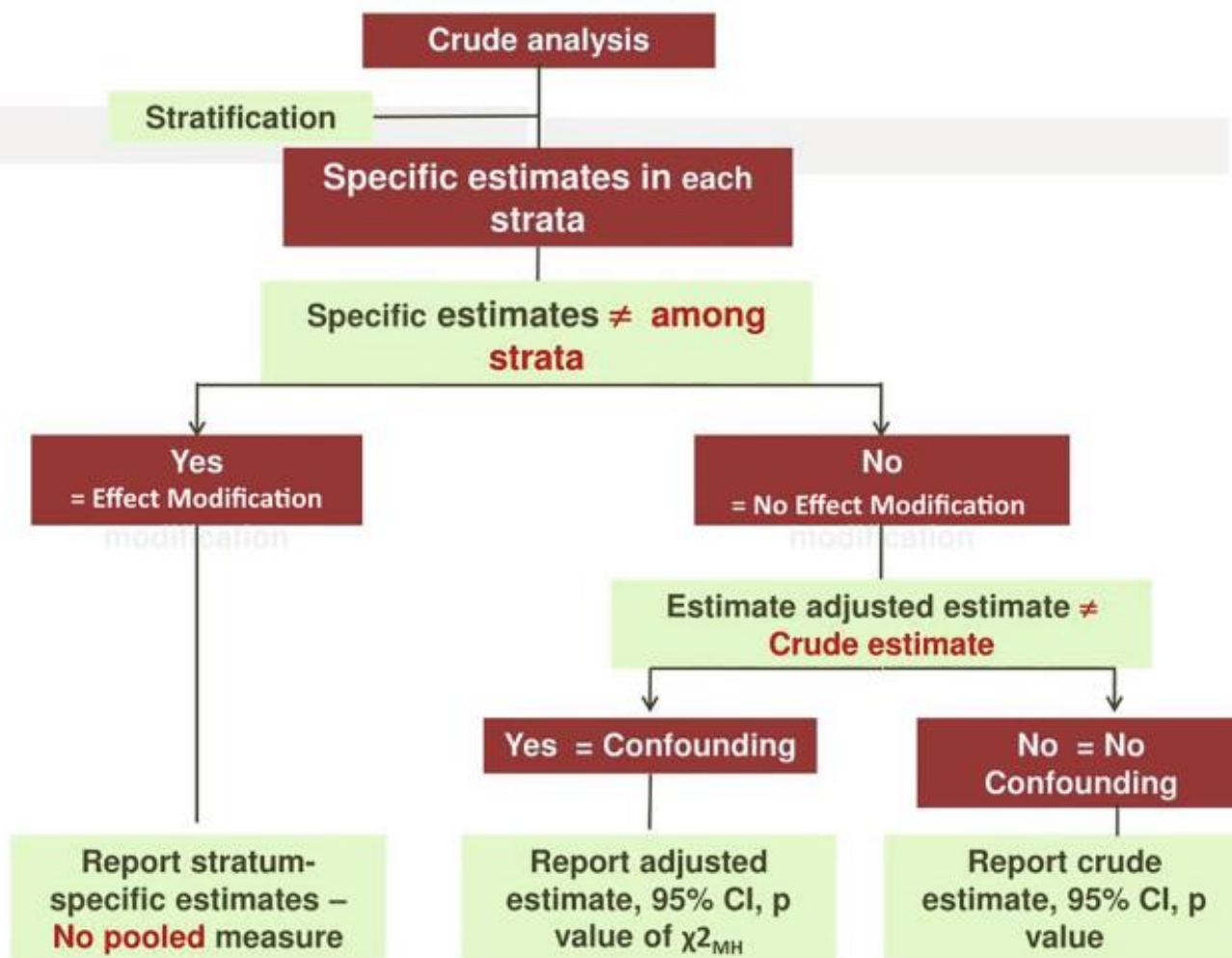


Repeated measure ANOVA Assumptions

- i. DV numerical and normally distributed
- ii. IV 2 or more groups
- iii. No significant outlier
- iv. Sphericity

Steps

- Analyze > General Linear Model > Repeated measure
- Enter number of time
- Enter the dependant variable
- Contrast – simple or repeated
- EM Means and post hoc - enter the main factor variable



HANDS-ON